PLD
Programmable Logic Devices

- 1. PROM (Programmable Read Only Memory)
- 2. PLA(Programmable Logic Array)
- 3. PAL(Programmable Array Logic)
- Logic Devices: (Adder, Subtractor, Multiplexer, Demultiplexer, Comparator)

Programmable Logic Array (PLA)

- Most flexible programmable device that consists of a programmable AND array and a programmable OR array

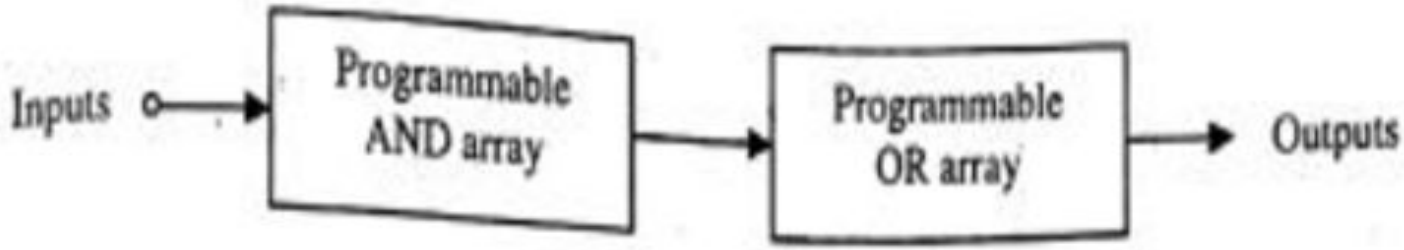

Fig.26: Block diagram of PLA architecture

To illustrate the basic concept, let us consider the following PLA architecture of Fig.27. A PLA is usually characterized by the number of inputs, the number of product terms, and the number of outputs. Accordingly, the following PLA has 4 inputs, 12 product terms, and 4 outputs.

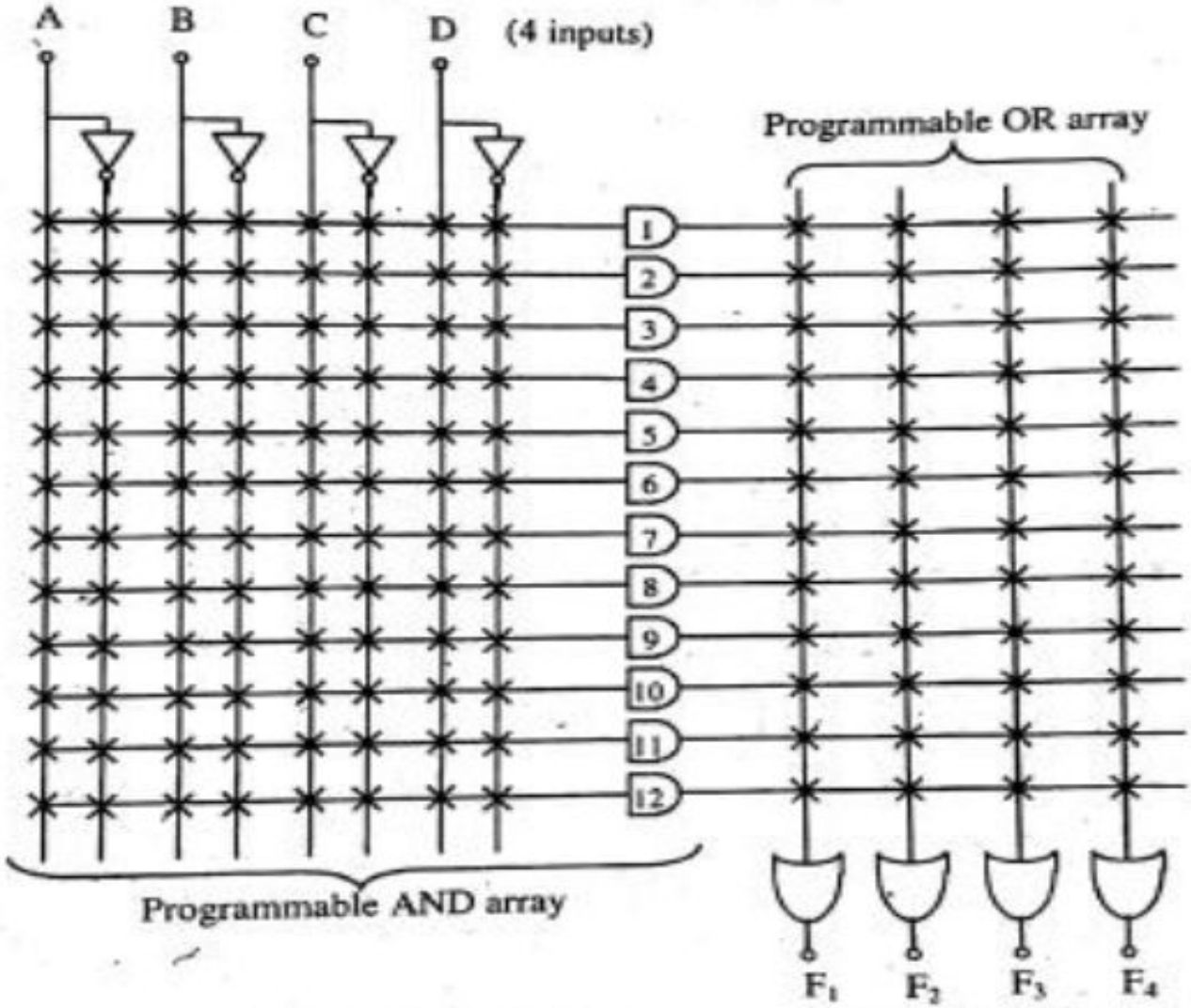

Fig.27: A 4×12×4 PLA architecture

This is the original configuration of a 4×12×4 PLA when produced by the manufacturers. Here each AND gate has all the 8 inputs (in complemented and uncomplemented form) with fusible links intact. Thus, each AND gate can be programmed to provide any desired product term. Similarly, each OR gate can be programmed to logically sum the product outputs of the AND gates, thereby implementing a sum-of-product expression. With this PLA, a designer can implement only 4 functions (F_1, F_2, F_3, and F_4) each having 12 product terms at best.

Example 6. Use a PLA to implement the following functions

$$F_1(A,B,C,D,E,F) = \overline{A} + D + \overline{B}CF$$

$$F_2(A,B,C,D,E,F) = BD\overline{F} + \overline{D}E + CE + \overline{C}DE\overline{F}$$

$$F_3(A,B,C,D,E,F) = \overline{C}D + AE + \overline{D}E$$

$$F_4(A,B,C,D,E,F) = \overline{C}D + \overline{B}\overline{D}F$$

Solution: At first simplify the logic functions using k-maps or some other methods as sum of products. Here the functions are already in simplified form as sum of products. Now observe that there are

i. 6 input variables,

ii. 12 product terms, and

iii. 4 output functions.

Clearly, a 6×12×4 PLA is sufficient to implement the functions. Such a 6×12×4 PLA architecture is shown below.

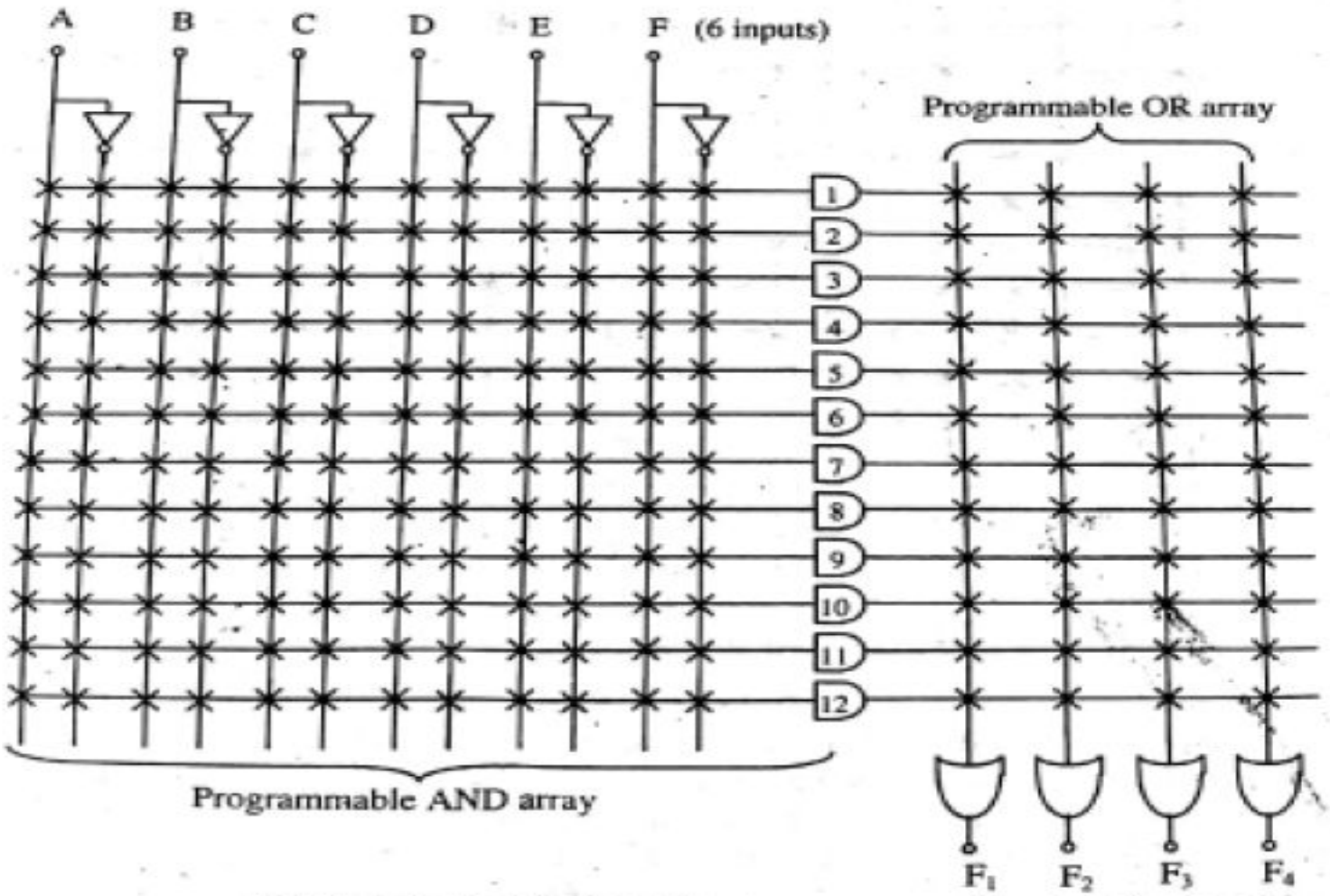

Fig.28: A 6×12×4 PLA architecture

This is the original configuration of a 6×12×4 PLA. Now program the AND array so that each AND gate provides one distinct product term. Similarly, program the OR array. The final implementation is shown in Fig.29.

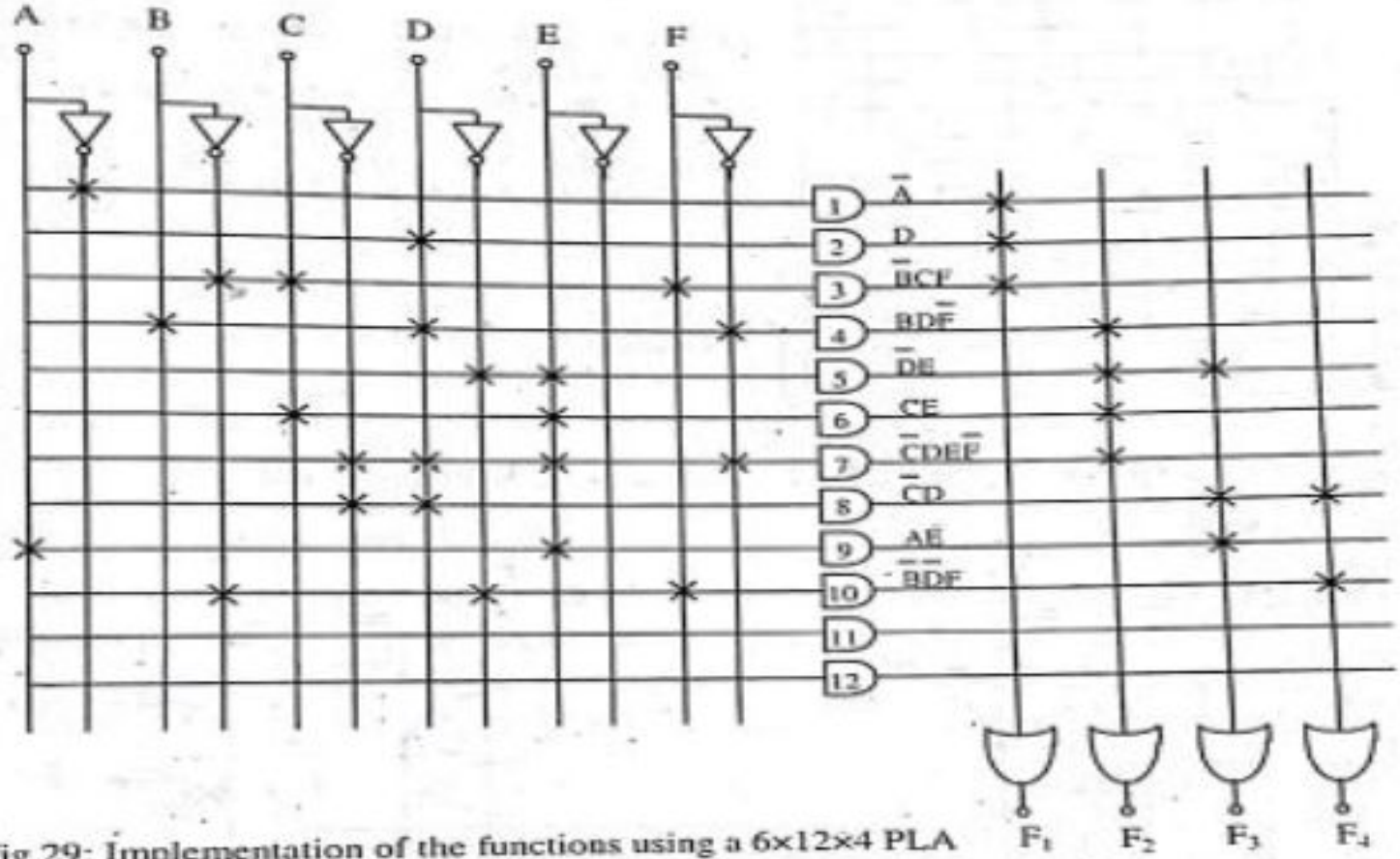

Fig.29: Implementation of the functions using a 6×12×4 PLA

Since there are 10 distinct product terms, AND gate-11 and -12 need not be programmed or connected. (Compare with example 5).

Ex: Design a BCD to Excees-3 code converter using PLA

BCD (input)				Excess-3 Code (Output)			
A	B	C	D	E3	E2	E1	E0
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0
1	0	1	0	x	x	x	x
1	0	1	1	x	x	x	x

- $E3=\sum m(5,6,7,8,9) + d(10,11,12,13,14,15)$
- $E2=\sum m(1,2,3,4,9) + d(10,11,12,13,14,15)$
- $E1=\sum m(0,3,4,7,8) + d(10,11,12,13,14,15)$
- $E0=\sum m(0,2,4,6,8) + d(10,11,12,13,14,15)$

From K-map,